

WASP-10b

R-1543

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_190929 · created 2026-08-01T19:20:27+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458755.7141 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.012
Transit depth (Rp/Rs)^2	2.23 +/- 0.36 [%]
Orbital Inclination (inc)	87.08 +/- 0.53
Airmass coefficient 1 (a1)	1.025 +/- 0.011
Airmass coefficient 2 (a2)	-0.022 +/- 0.02
Scatter in the residuals of the lightcurve fit is	1.05 %
Best Comparison Star	#2 - [543, 111]
Optimal Method	PSF photometry
Transit Duration (day)	0.0809 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.149 ± 0.012 vs 0.1592 (deviation 0.85σ)
Duration fit vs published	0.0809 d vs 0.0946 d (deviation 2.58σ)
Mid-time O-C	-4.5 min
Depth SNR	12.4
Residual scatter	1.05 %

Light curve

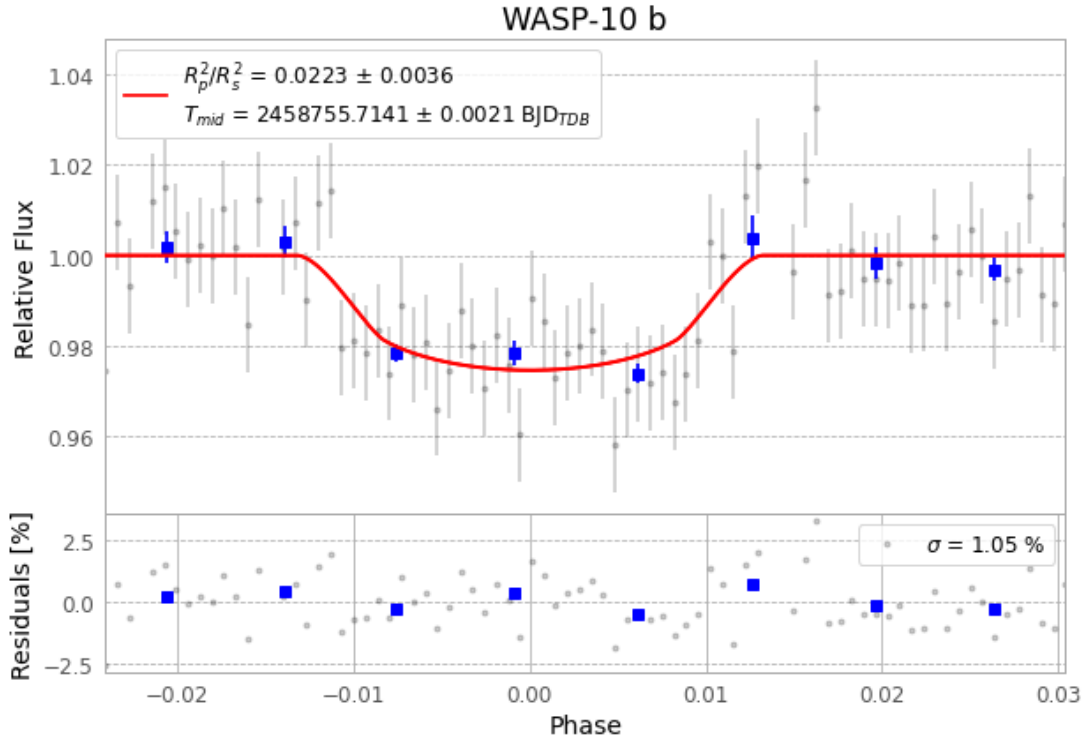


Figure 1 — FinalLightCurve_WASP-10 b_2019-09-28.png (SHA-256 886eeae4b7e97110...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [326, 189], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 97fee2a4447793ff...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

81 FITS frames (53 MB), WASP-10190929031212.FITS → WASP-10190929071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-190929.log (584edd8fa3cd...), FinalLightCurve_WASP-10 b_2019-09-28.png (886eeae4b7e9...), FinalLightCurve_WASP-10 b_2019-09-28.csv (4705bdeb00d5...)

Compute resources

Wall time 175.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 19:20Z.